

Lab: Derivation Exercise — Coupled Inference and Multi-Agent Free Energy

Objective

Derive the mathematics of communication between Active Inference agents, from coupled generative models to generalized synchrony conditions.

Part 1: Two-Agent Generative Models

Goal: Formalize how two agents model each other.

Setup: Agent A and Agent B, each with 2 states. Agent A's generative model:

- $p_A(o_A | s_A, s_B)$ — likelihood depends on both agents' states
- $p_A(s_A) = N(s_A; 0, 1)$ — prior over own states
- $p_A(s_B) = N(s_B; 0, \sigma^2_B)$ — prior over B's states (higher uncertainty)

Agent B's generative model is symmetric.

Task:

1. Write A's free energy F_A as a function of $q_A(s_A)$ and $q_A(s_B)$
2. Derive the update equations for A's beliefs about s_A and s_B
3. Show that A's beliefs about B depend on A's observations (which are influenced by B's actions)

Write your response here

Part 2: Joint Free Energy Decomposition

Goal: Derive the relationship between individual and joint free energy.

Task:

1. Write the joint free energy: $F_{\text{joint}} = F_A + F_B - I(s_A; s_B)$
2. Show that the mutual information $I(s_A; s_B) = E[\ln q(s_A, s_B) / (q(s_A)q(s_B))]$ captures the coupling
3. Explain why minimizing F_{joint} simultaneously minimizes individual free energies and maximizes mutual predictability
4. Interpret: when $I(s_A; s_B)$ is high, the agents are in tight communication; when it is low, they are operating independently

Write your response here

Part 3: Generalized Synchrony Conditions

Goal: Derive the conditions for synchronization.

Setup: Two coupled dynamical systems:

- $dx_1/dt = f_1(x_1) + \kappa \cdot (x_2 - x_1)$
- $dx_2/dt = f_2(x_2) + \kappa \cdot (x_1 - x_2)$

where κ is the coupling strength.

Task:

1. Define the synchronization error $e = x_1 - x_2$
2. Derive the dynamics of e : $de/dt = f_1(x_1) - f_2(x_2) - 2\kappa \cdot e$
3. Show that for identical systems ($f_1 = f_2 = f$), $e \rightarrow 0$ is stable if $\kappa > \lambda_{\text{max}}/2$, where λ_{max} is the maximum Lyapunov exponent of f

4. Interpret: coupling must be strong enough relative to the internal dynamics for synchronization to occur

Write your response here

Part 4: Theory of Mind as Hierarchical Modeling

Goal: Formalize Theory of Mind mathematically.

Setup: Agent A's hierarchical model:

- Level 1: $p_A(o_A | s_A)$ — how A's states generate A's observations
- Level 2: $p_A(s_A | s_B \text{ model})$ — how B's model of the world influences A's states
- Level 3: $p_A(s_B \text{ model} | s_B \text{ meta-model})$ — A's model of B's model of A

Task:

1. Write the free energy for this three-level hierarchy
2. Show how increasing hierarchical depth enables more sophisticated social inference
3. Compute the additional cost (complexity) of each added level
4. Discuss: what is the optimal depth of Theory of Mind modeling?

Write your response here

Part 5: Synthesis

In 200 words, explain how the mathematics of coupled inference, joint free energy, and generalized synchrony provide a unified account of communication — from simple biological coordination to human language.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Multi-agent modeling	Two-agent free energy formulation
2	Information theory	Joint free energy decomposition
3	Dynamical systems	Synchronization stability conditions
4	Hierarchical modeling	Theory of Mind as recursive generative models
5	Conceptual integration	Connecting mathematical communication to natural language